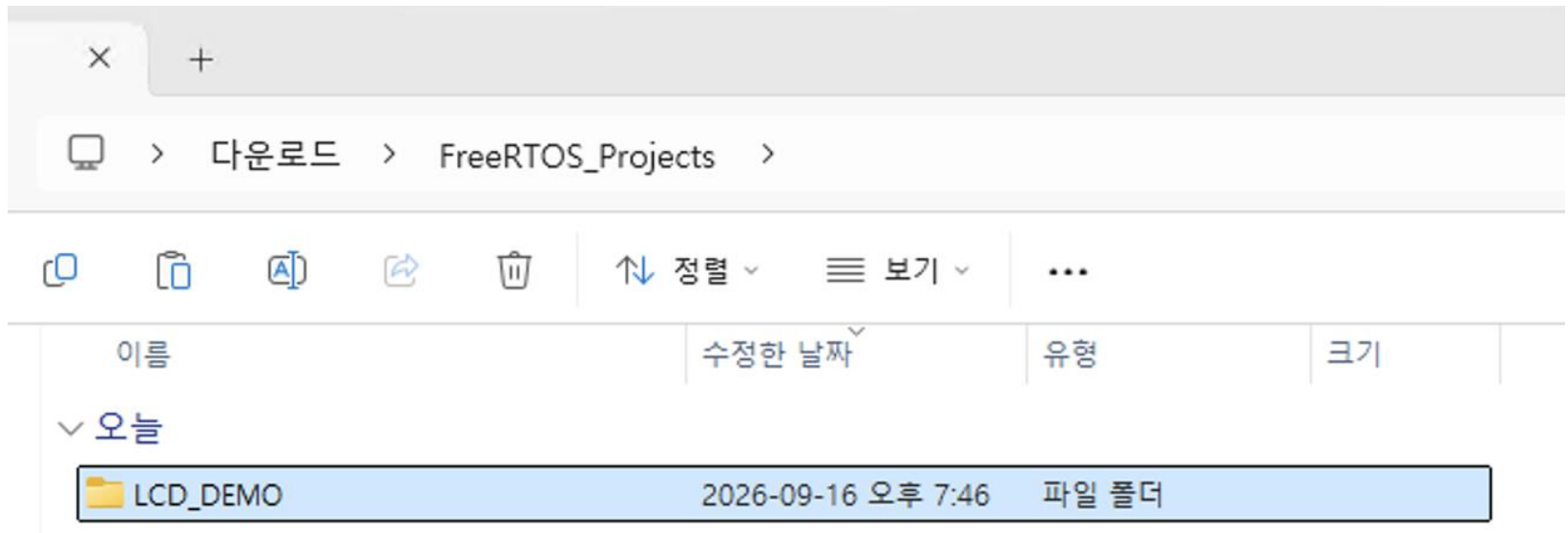


LCD Demo 실습

LCD Demo 실습

- FreeRTOS_Projects 디렉터리 내
 - LCD_DEMO 디렉터리 생성

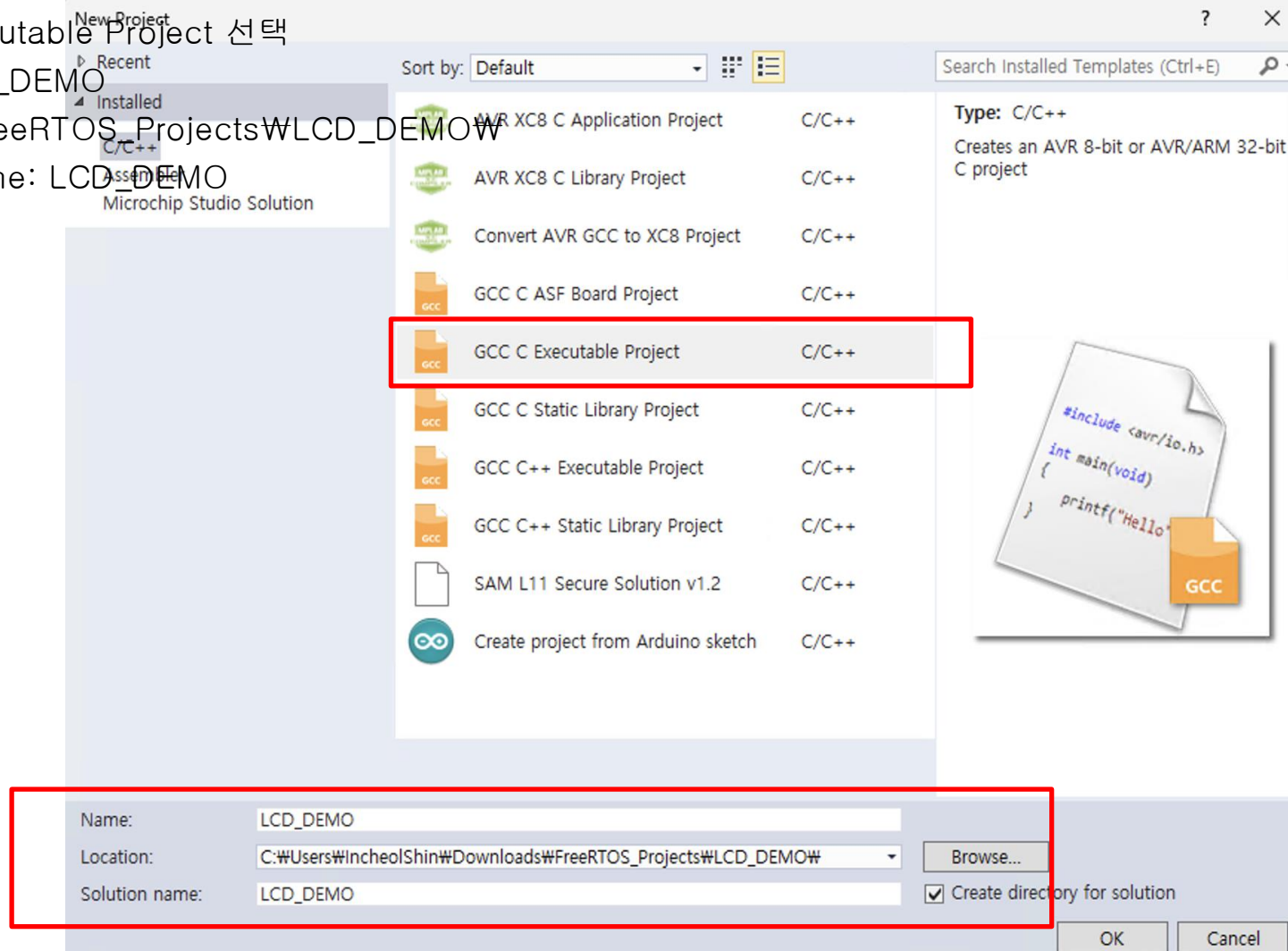


LCD Demo 실습

□ LCD_DEMO 프로젝트 생성

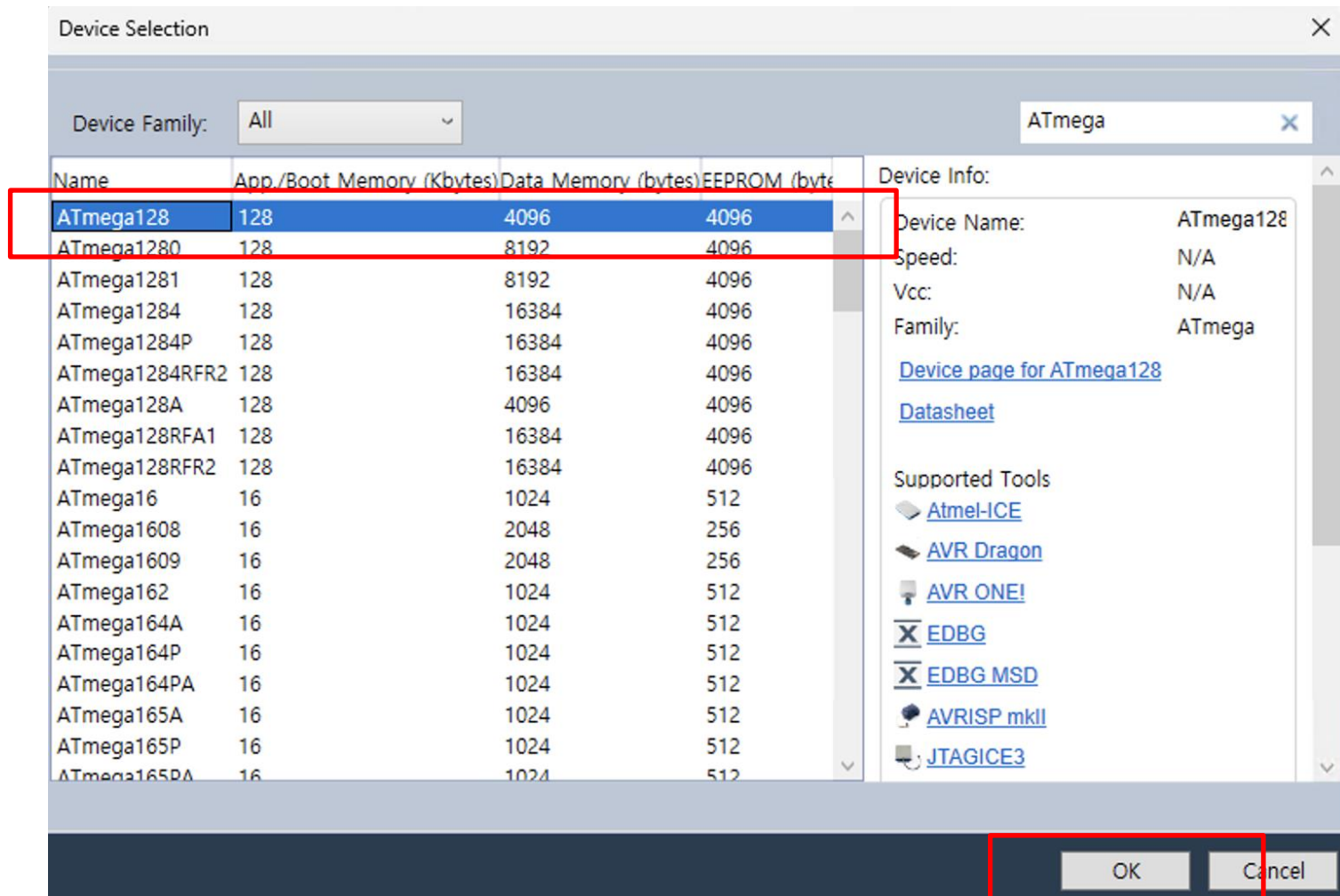
○ Microchip Studio 프로그램 실행 후

- GCC C Executable Project 선택
- Name : LCD_DEMO
- Location: FreeRTOS_Projects\LCD_DEMO\
- Solution name: LCD_DEMO



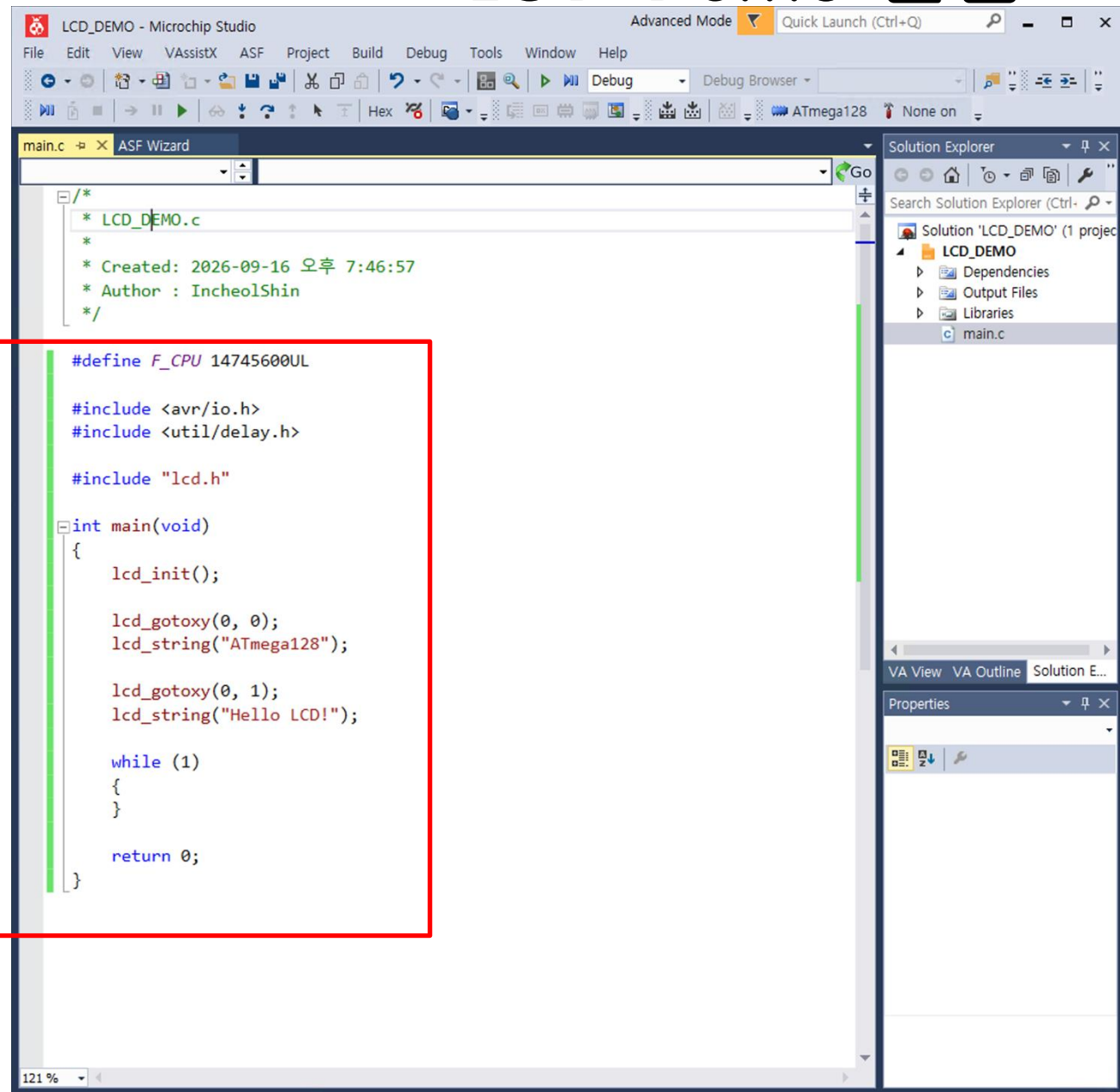
LCD Demo 실습

- LCD_DEMO 프로젝트 생성 (계속)
- Microchip Studio 프로그램 실행 후 (계속)
 - ATmega128 선택 후 'OK'



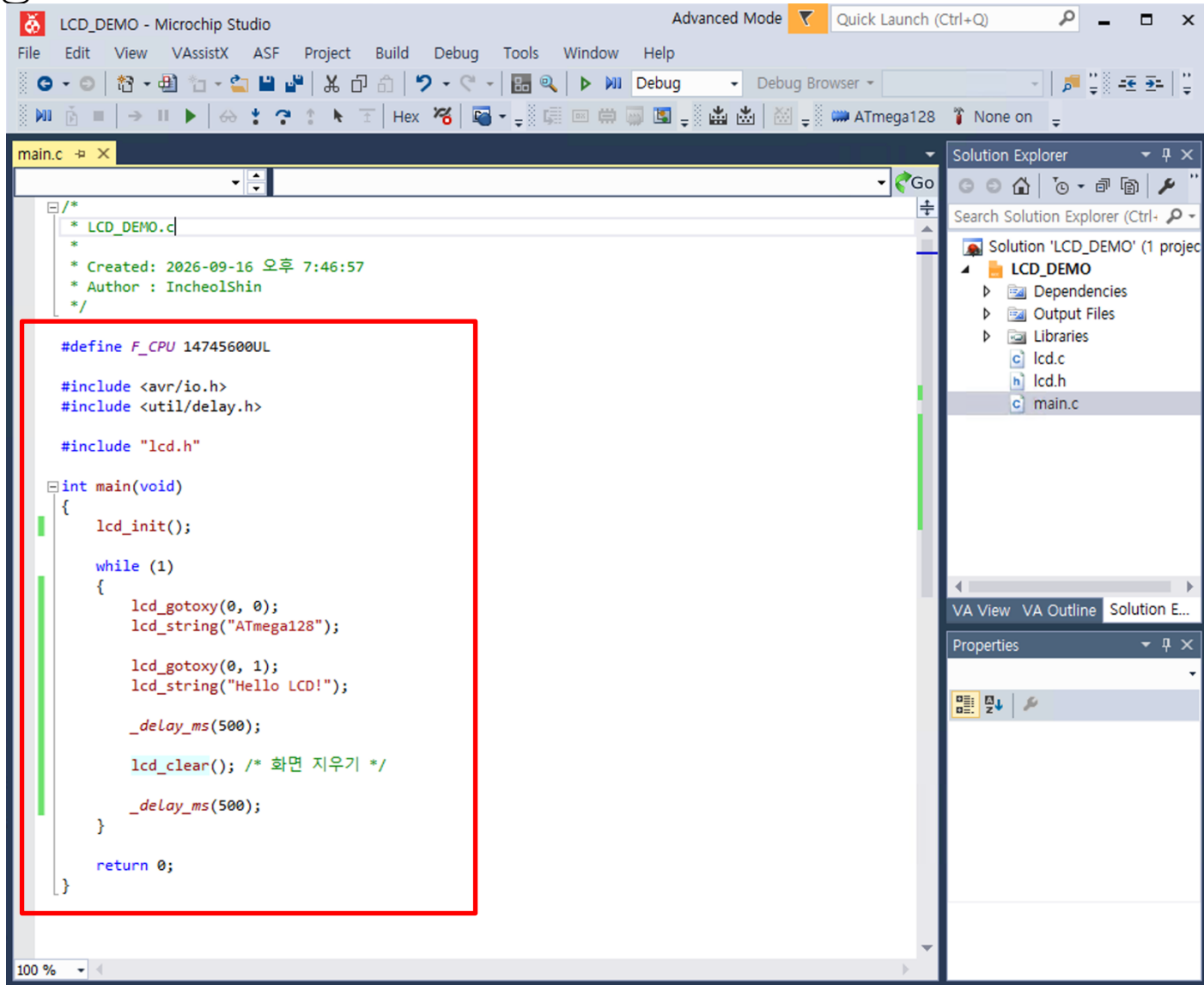
LCD Demo 실습

□ 'main' 함수 작성



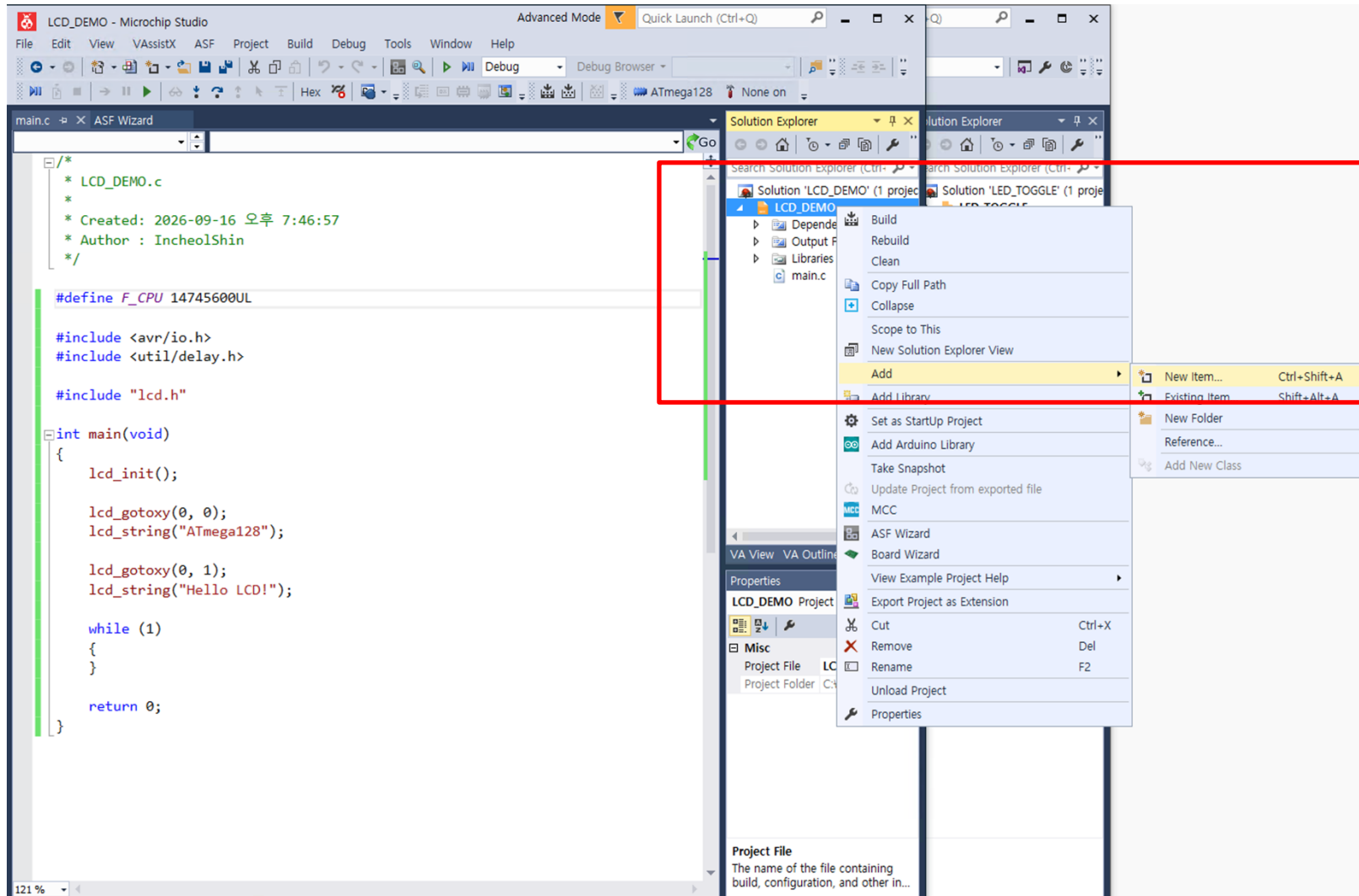
LCD Demo 실습

□ 'main' 함수 작성



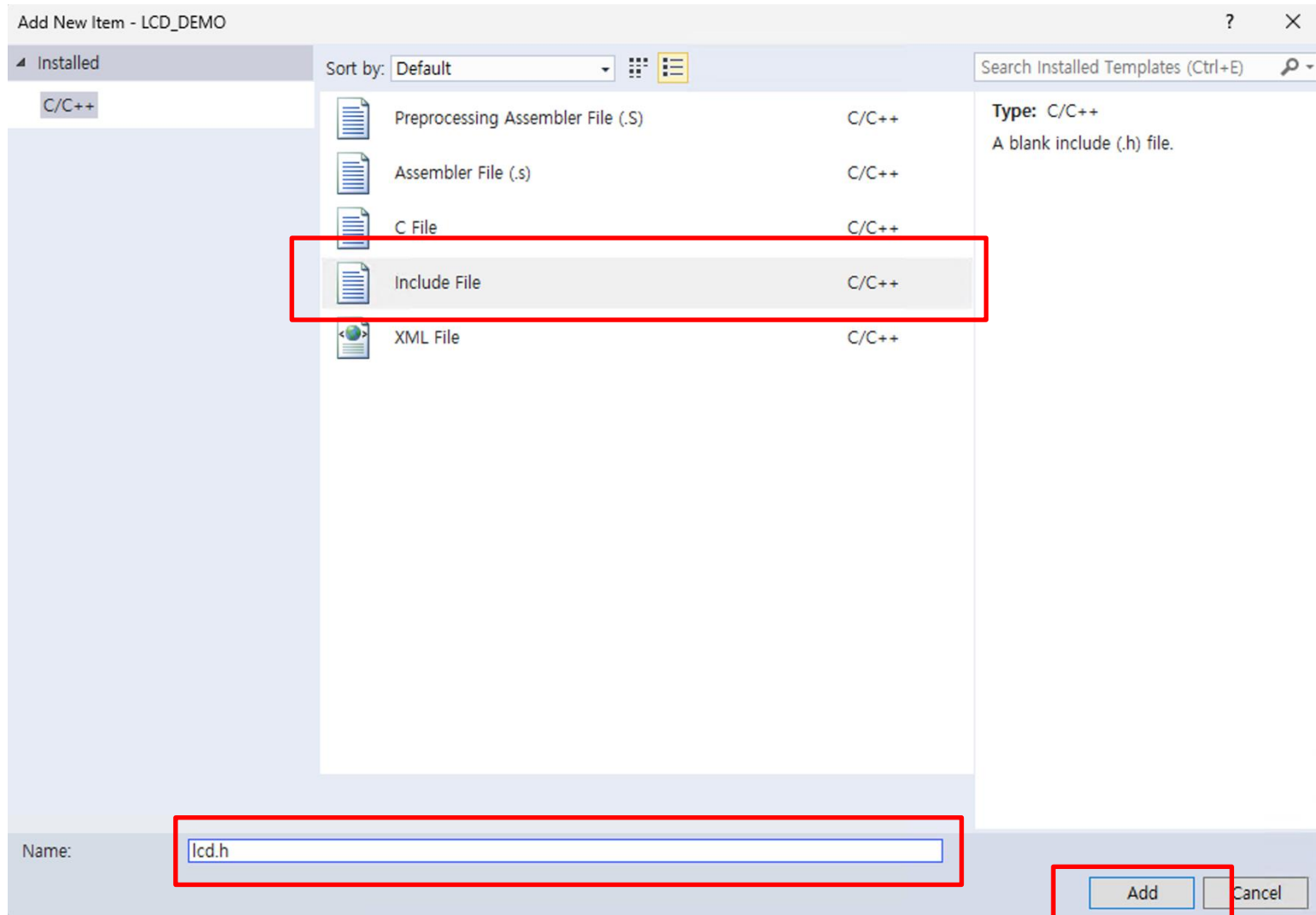
LCD Demo 실습

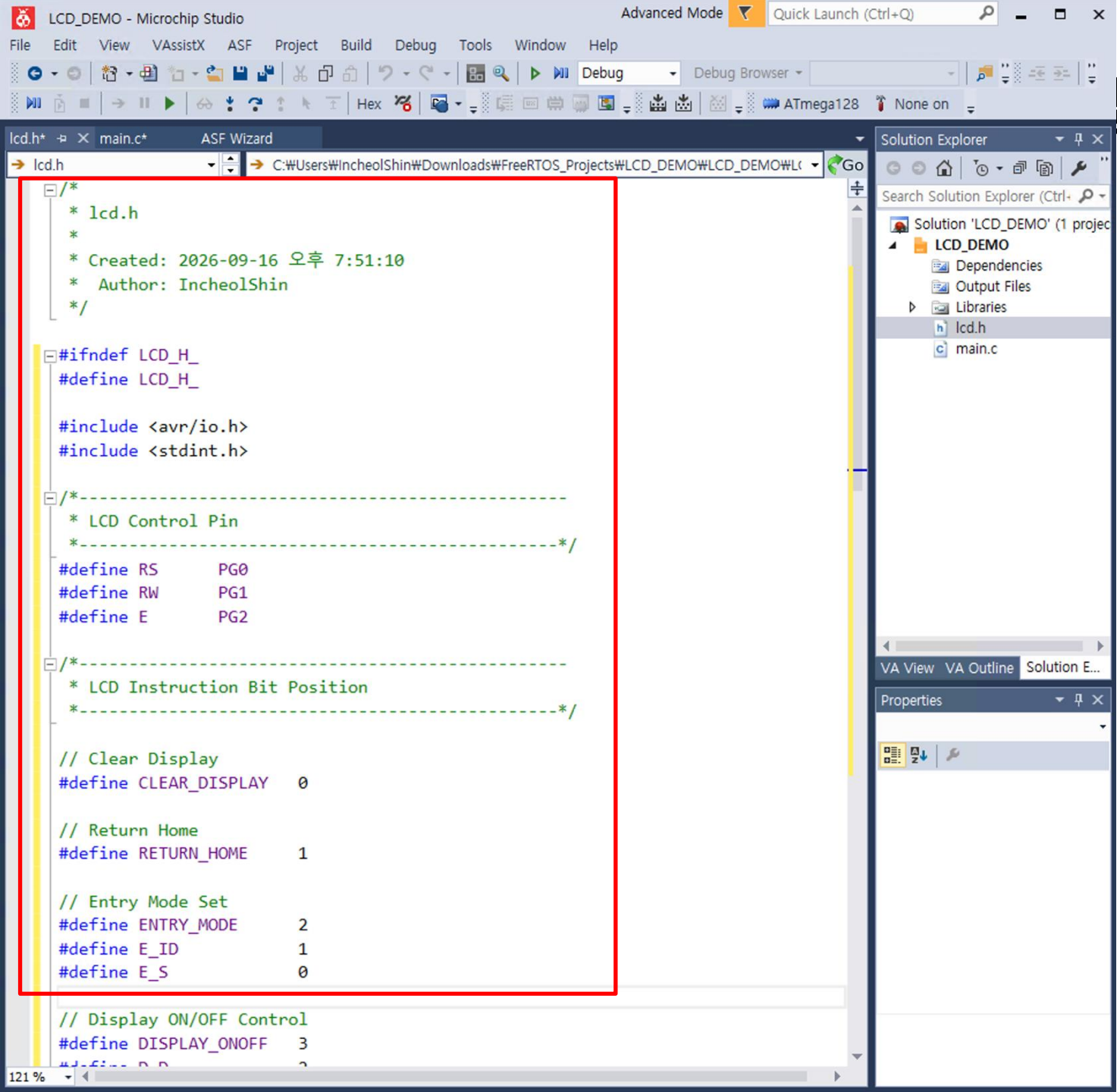
- 'lcd.h' 헤더파일 작성
- 프로젝트에 'lcd.h' 헤더파일 추가

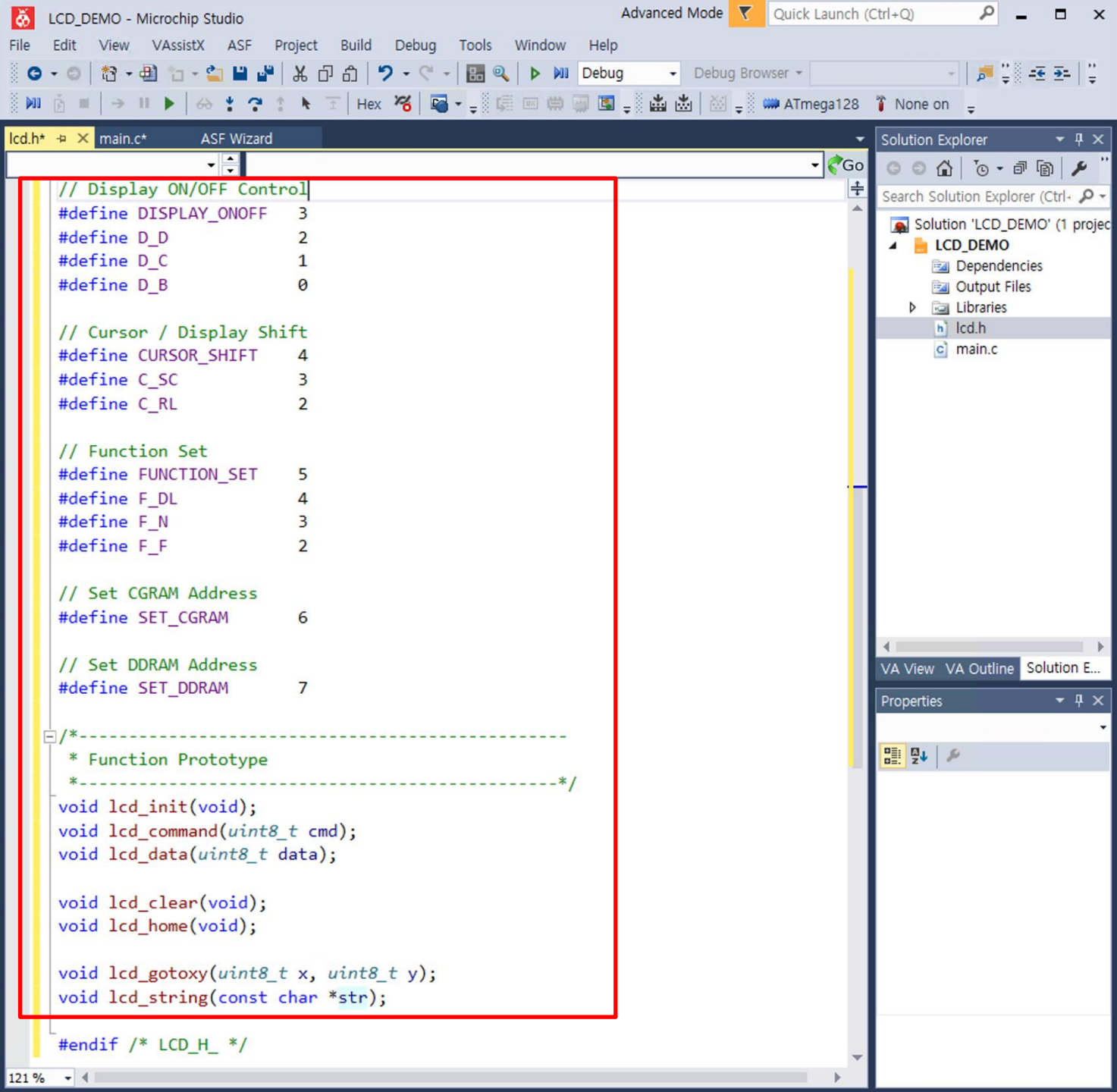


LCD Demo 실습

- 'lcd.h' 헤더파일 작성 (계속)
- 프로젝트에 'lcd.h' 헤더파일 추가 (계속)



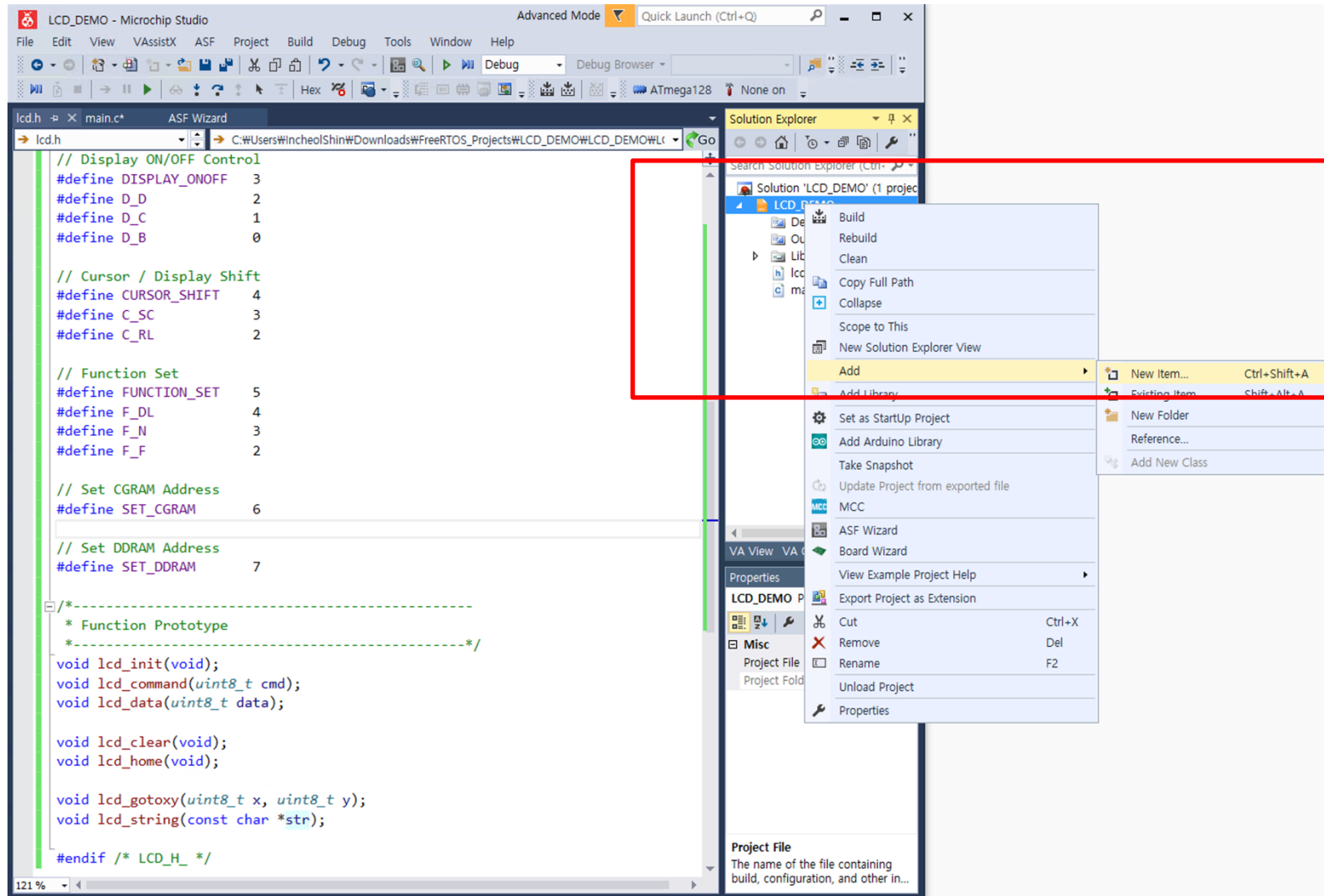




LCD Demo 실습

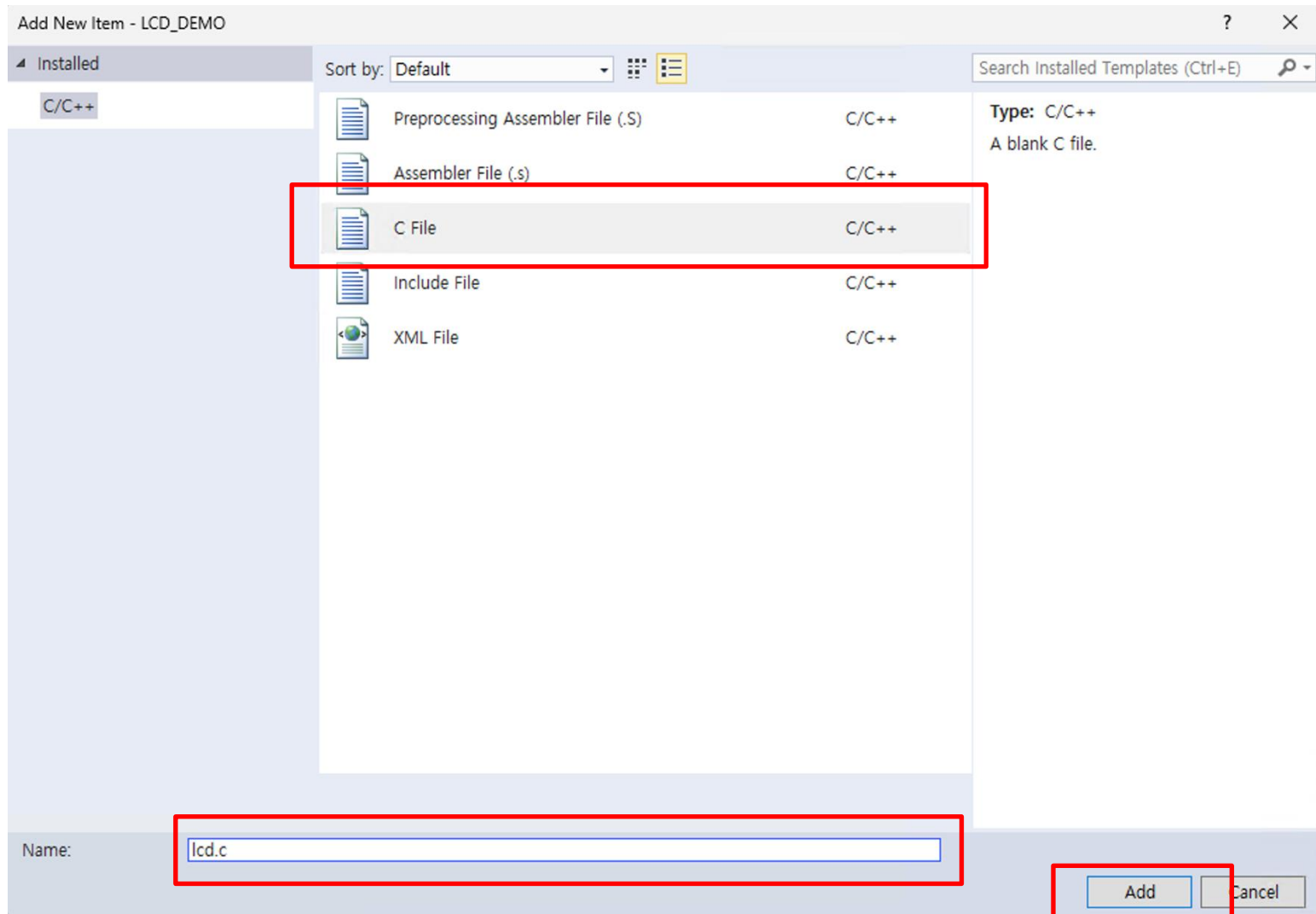
□ 'lcd.c' 소스파일 작성

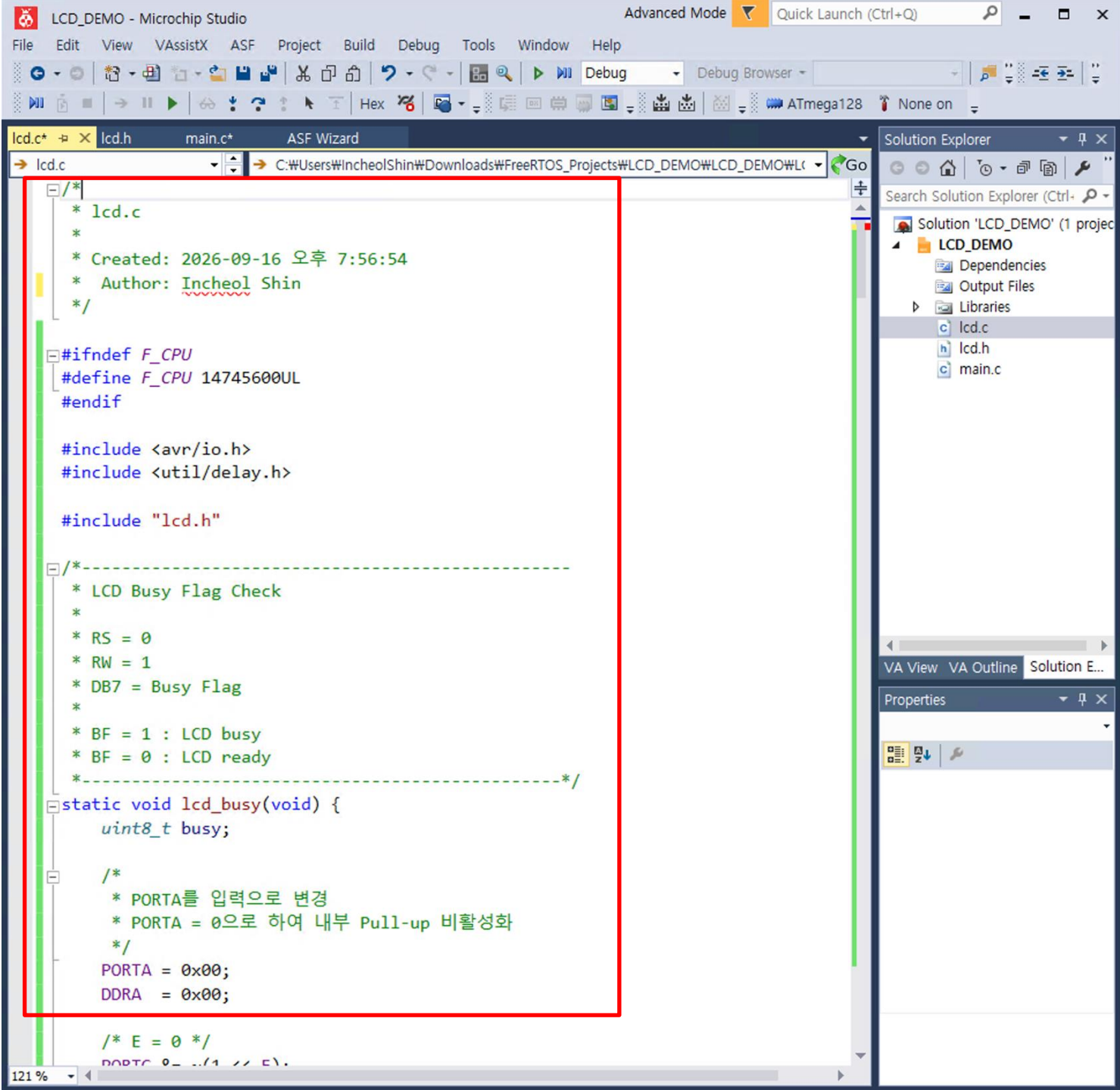
○ 프로젝트에 'lcd.c' 소스파일 추가

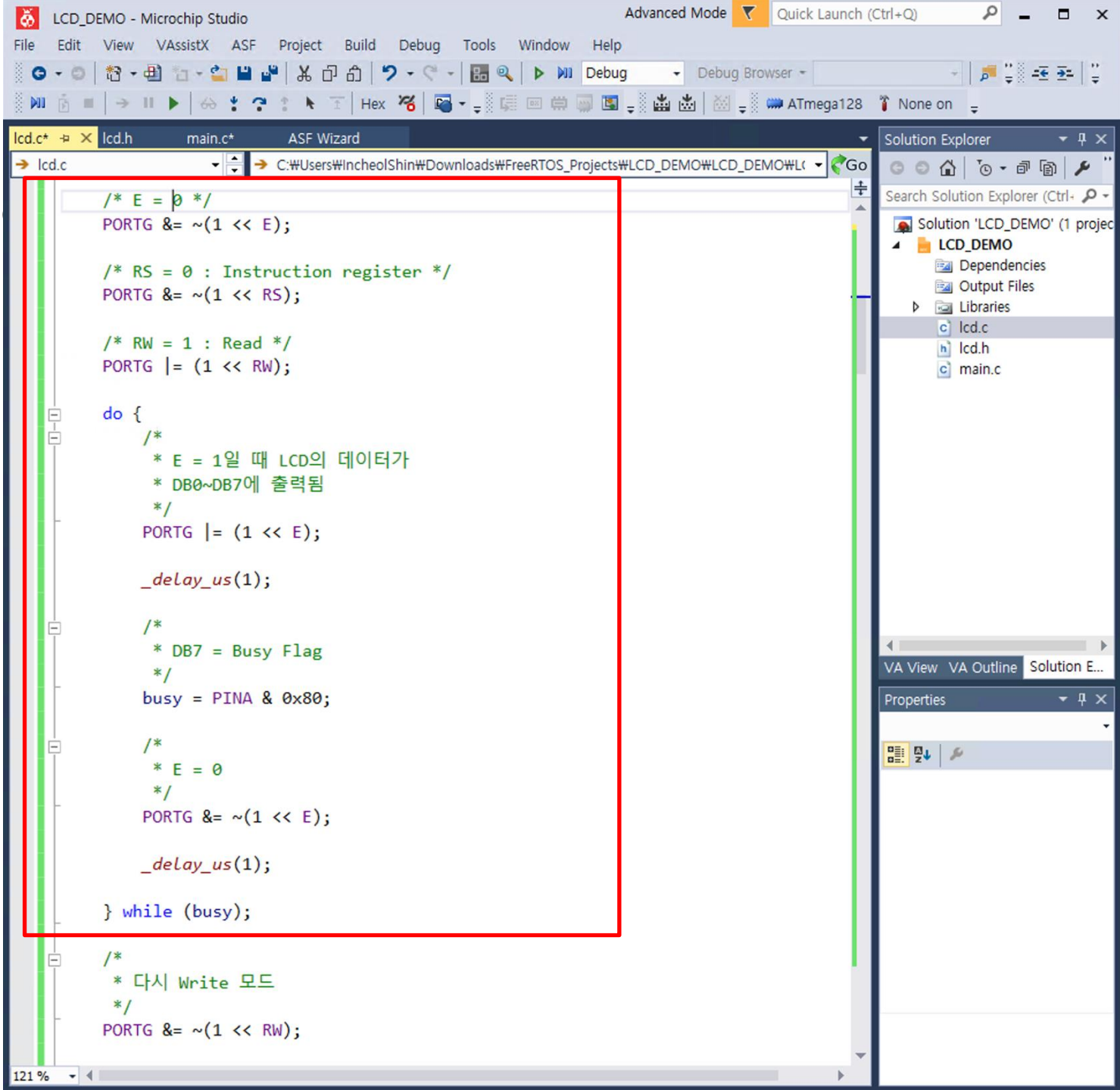


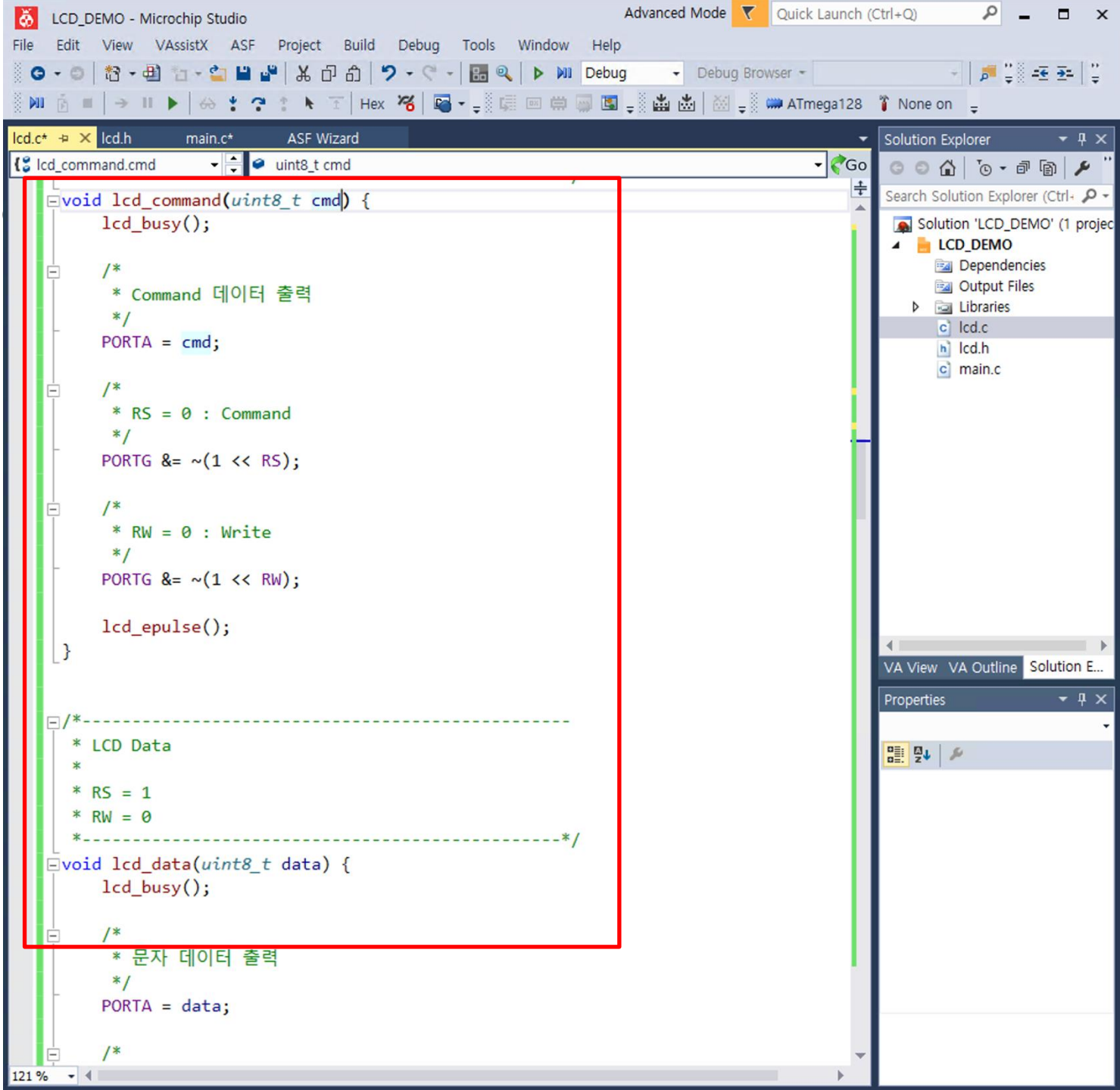
LCD Demo 실습

- 'lcd.c' 소스파일 작성 (계속)
- 프로젝트에 'lcd.c' 소스파일 추가 (계속)

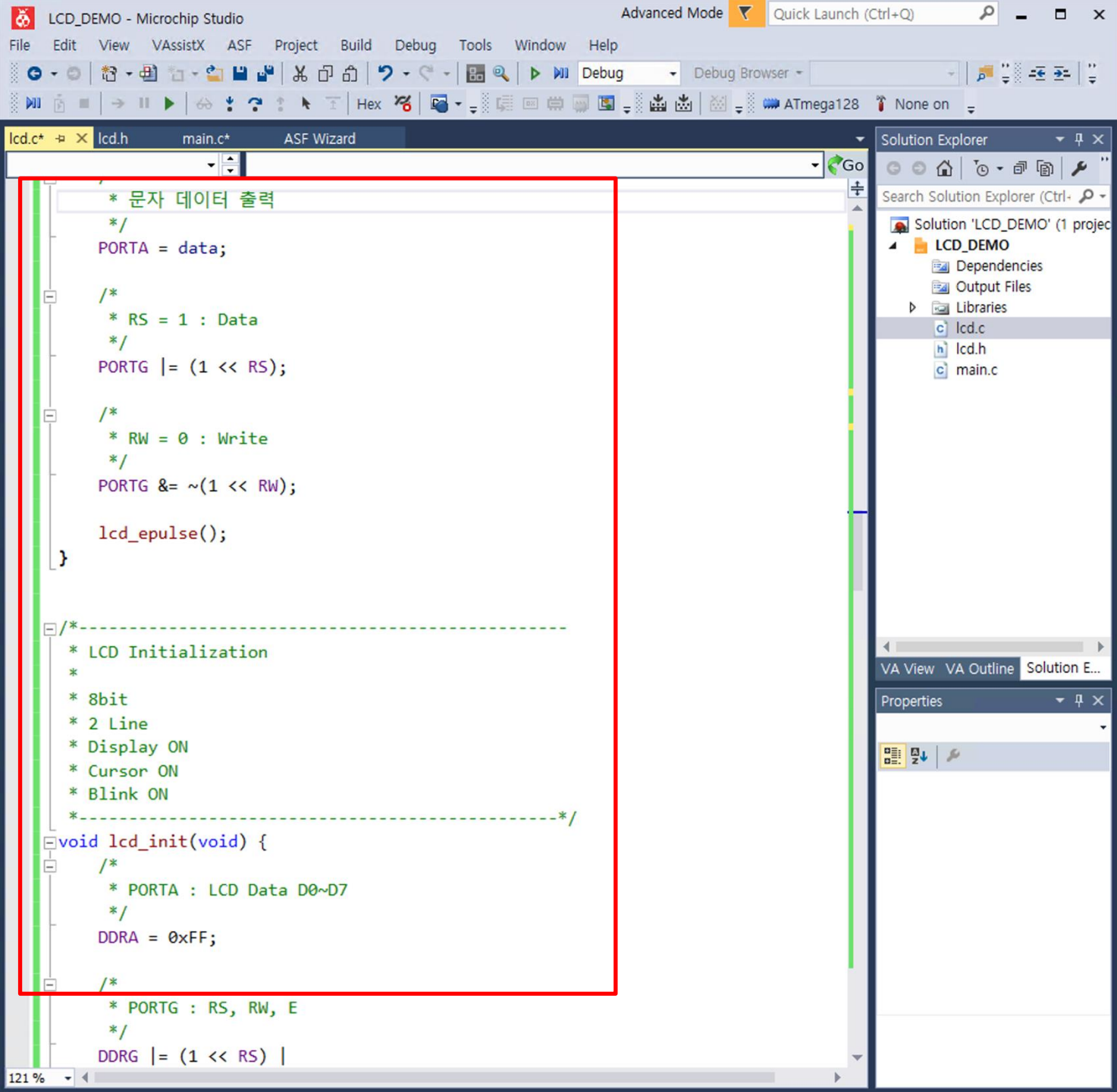


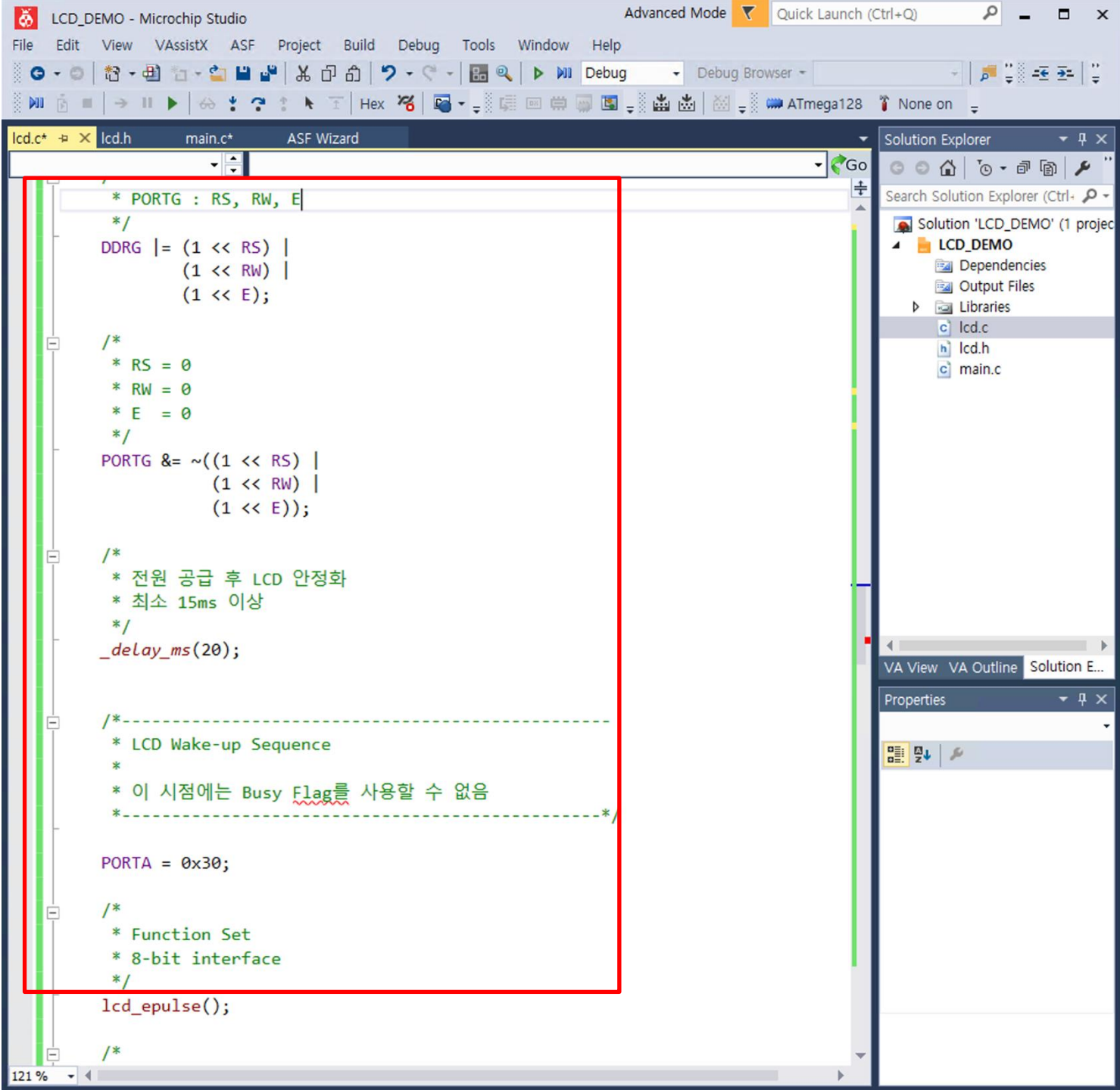




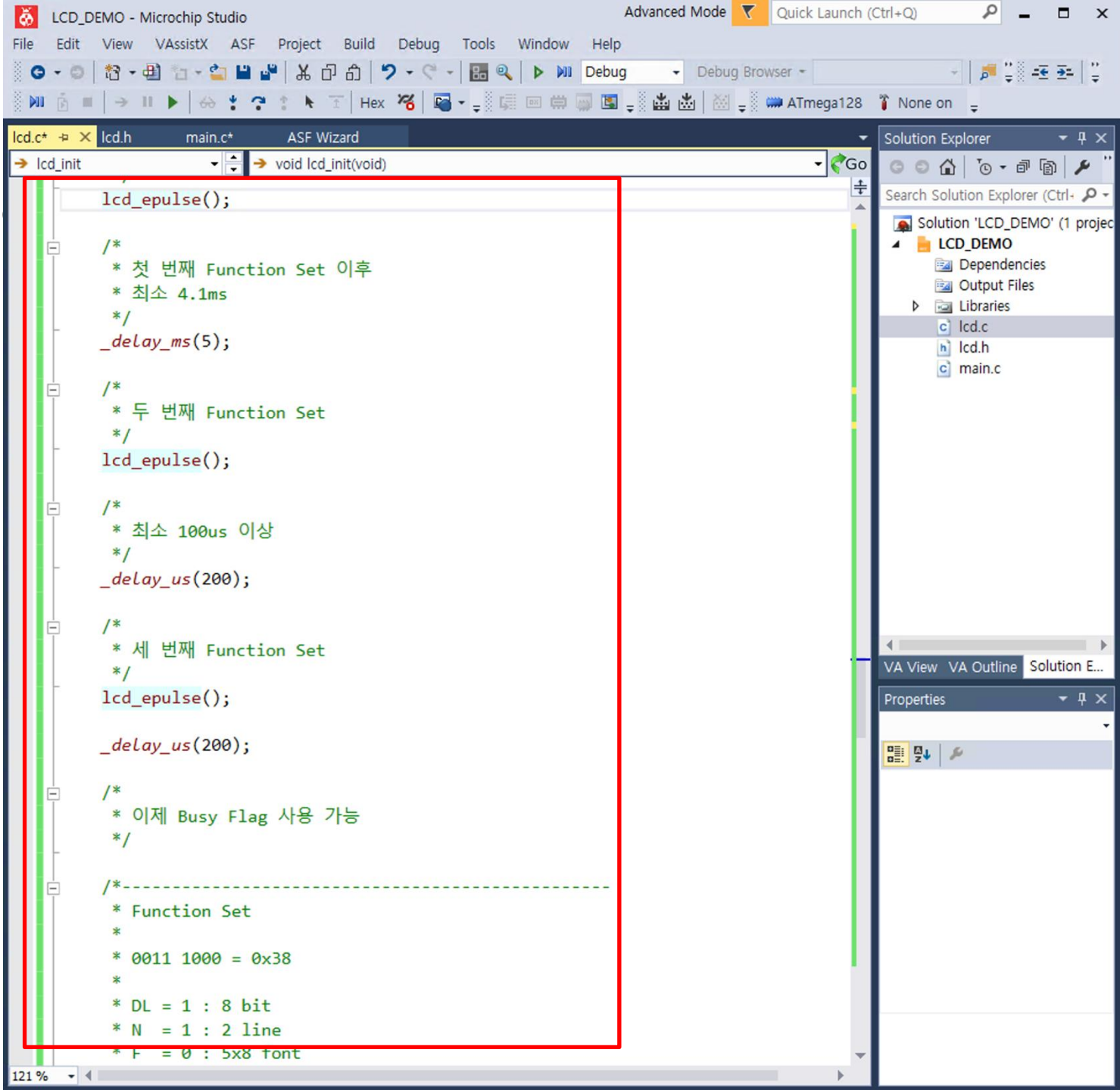


승

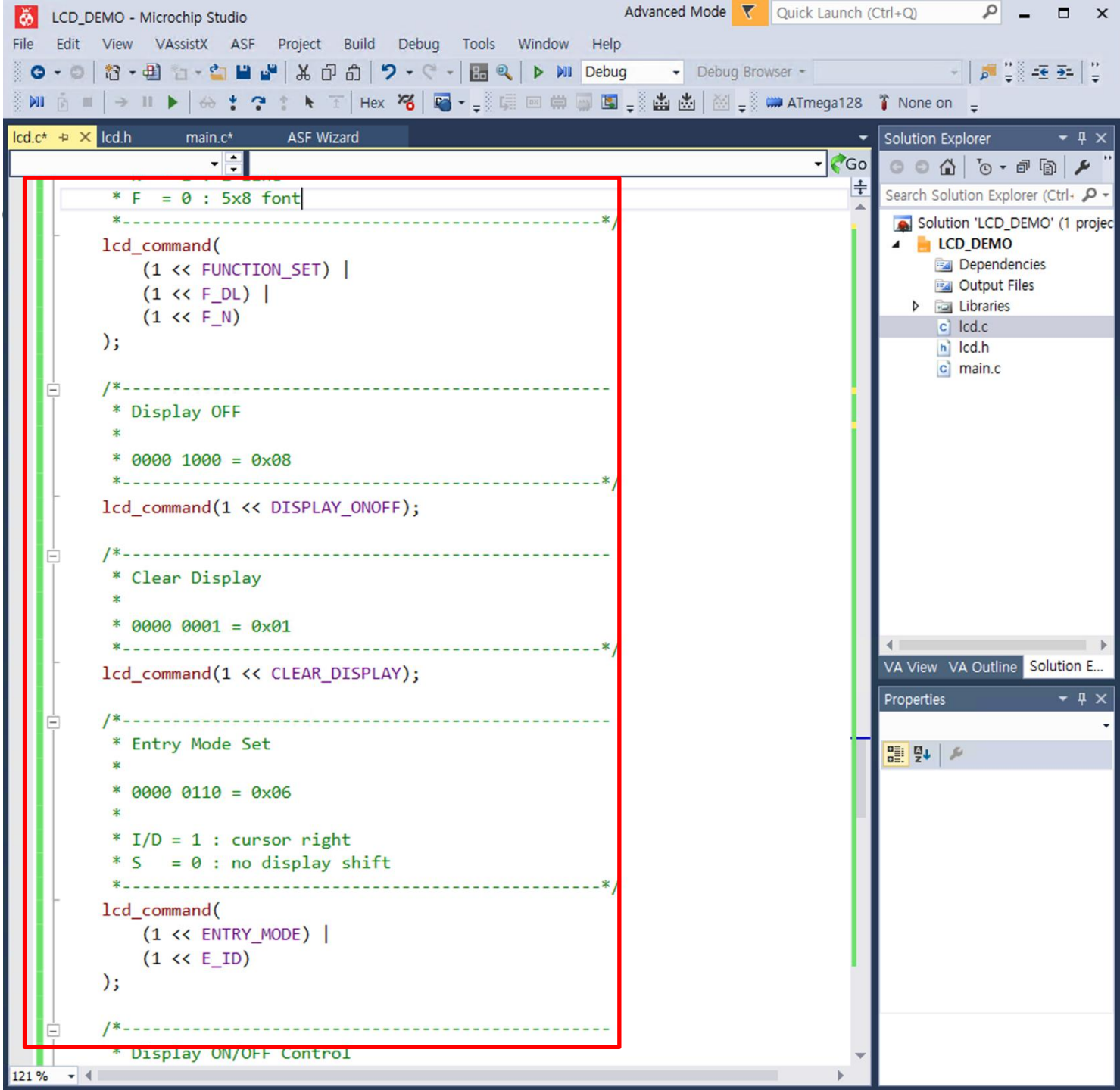


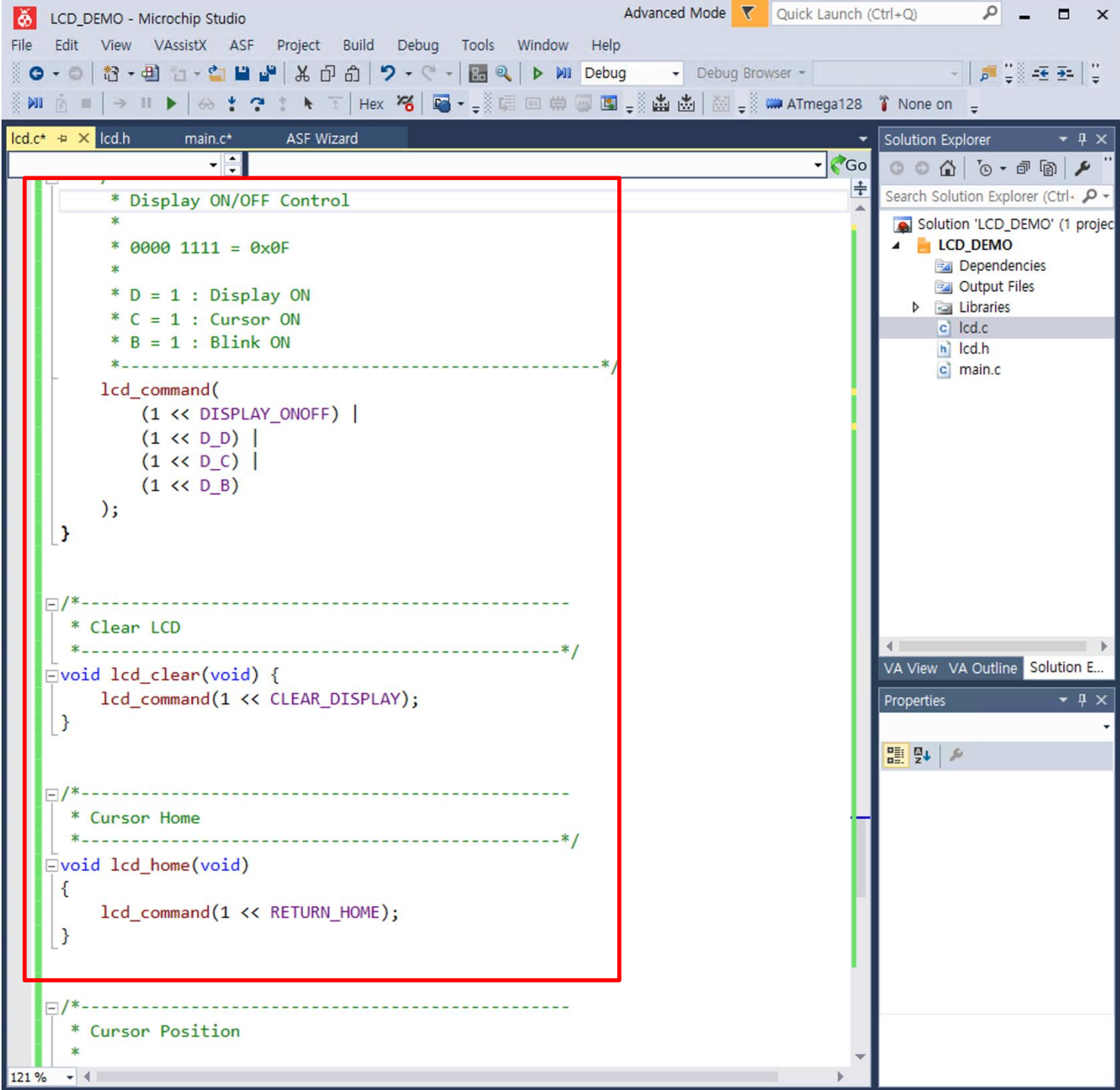


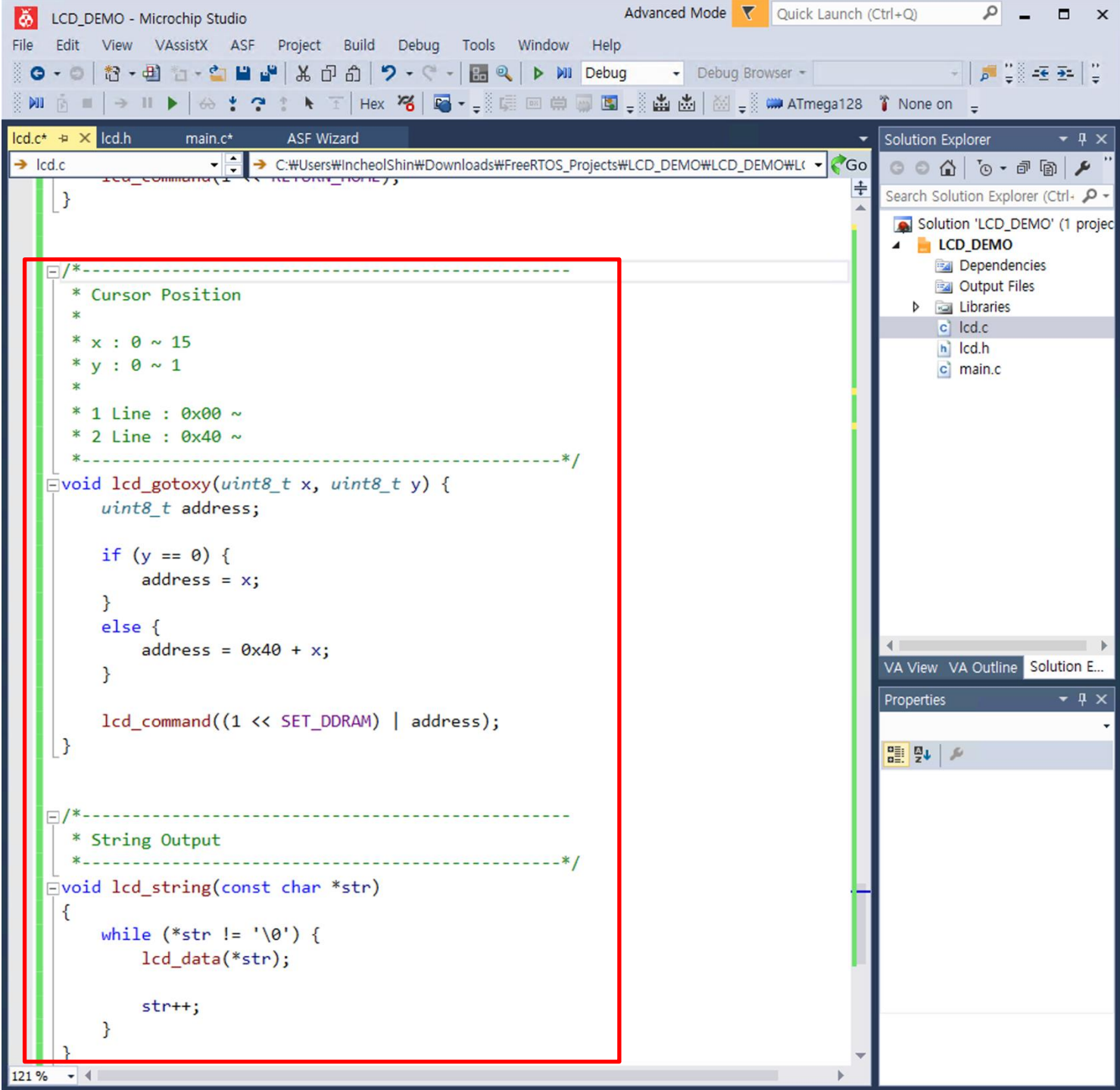
승



승

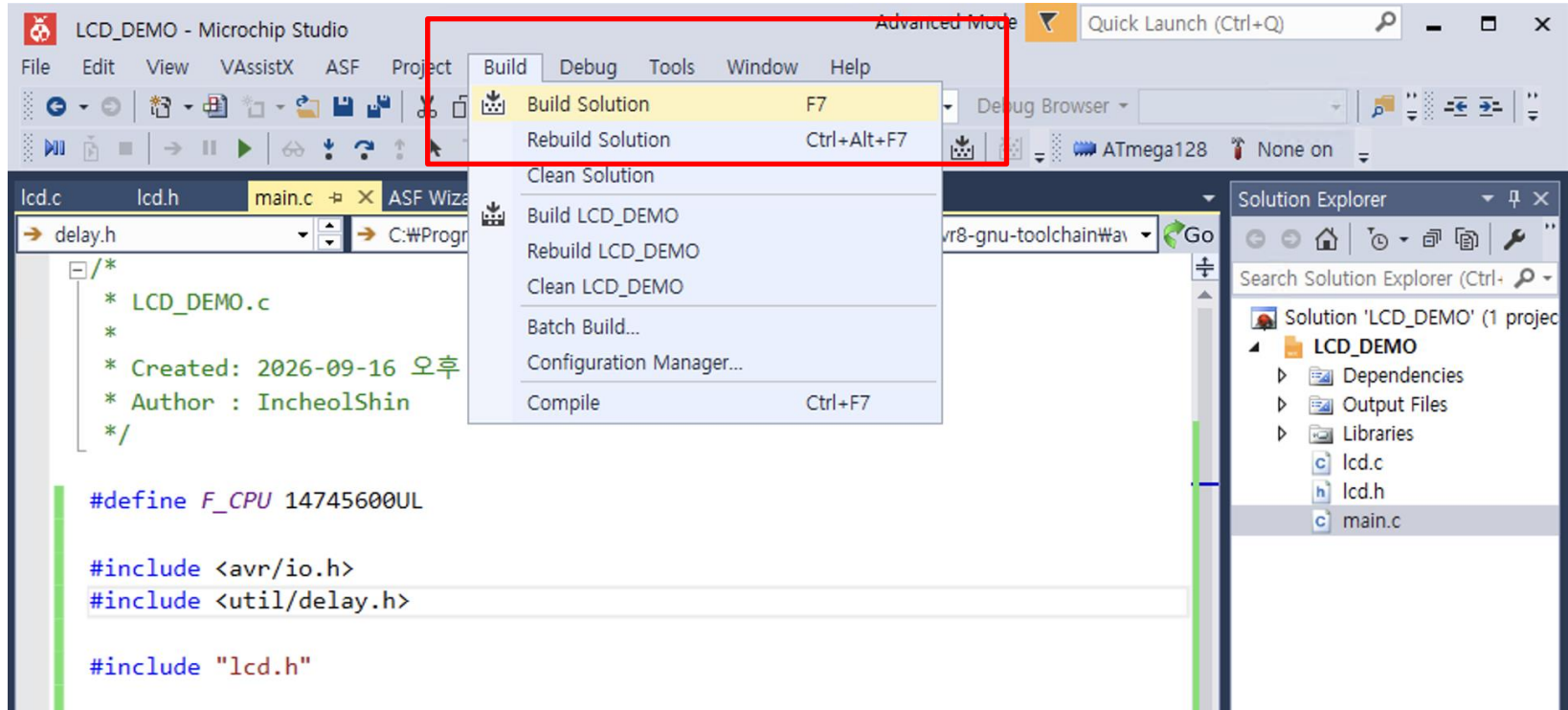






LCD Demo 실습

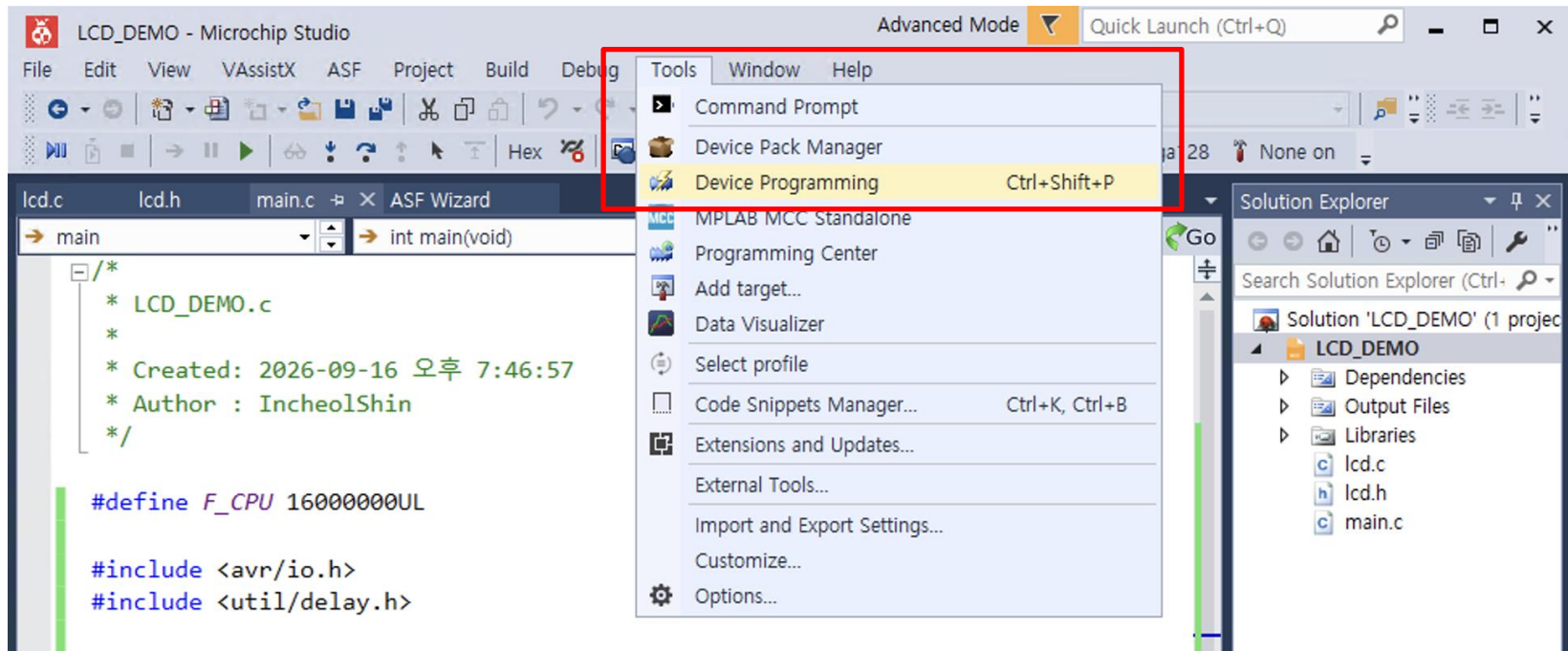
□ Build Solution



LCD Demo 실습

□ Device Programming

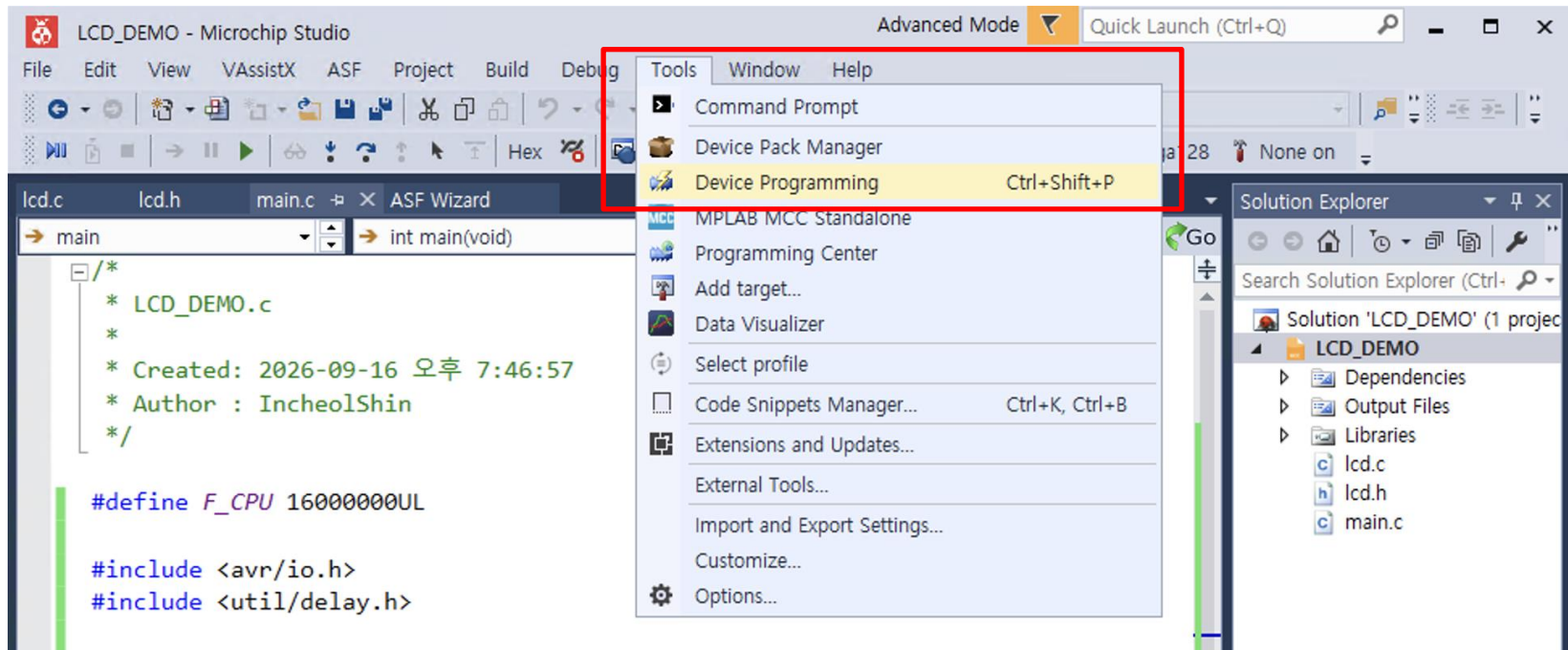
- 실행 파일을 임베디드 보드에 업로드



LCD Demo 실습

□ Device Programing (계속)

○ 실행



LCD Demo 실행

AVRISP mkII (000200212345) - Device Programming

Tool: AVRISP mkII Device: ATmega128 Interface: ISP Device signature: 0x1E9702 Target Voltage: 3.3 V

Interface settings
Tool information
Device information
Oscillator calibration
Memories
Fuses
Lock bits
Production file

Device
Erase Chip Erase now

Flash (128 KB)
C:\Users\WincheolShin\Downloads\FreeRTOS_Projects\LCD_DEMO\LCD_DEMO\LCD_DEMO...
☒ Erase device before programming
☒ Verify Flash after programming
Program Verify Read...
Advanced

EEPROM (4 KB)
...
☒ Verify EEPROM after programming
Program Verify Read...
Advanced

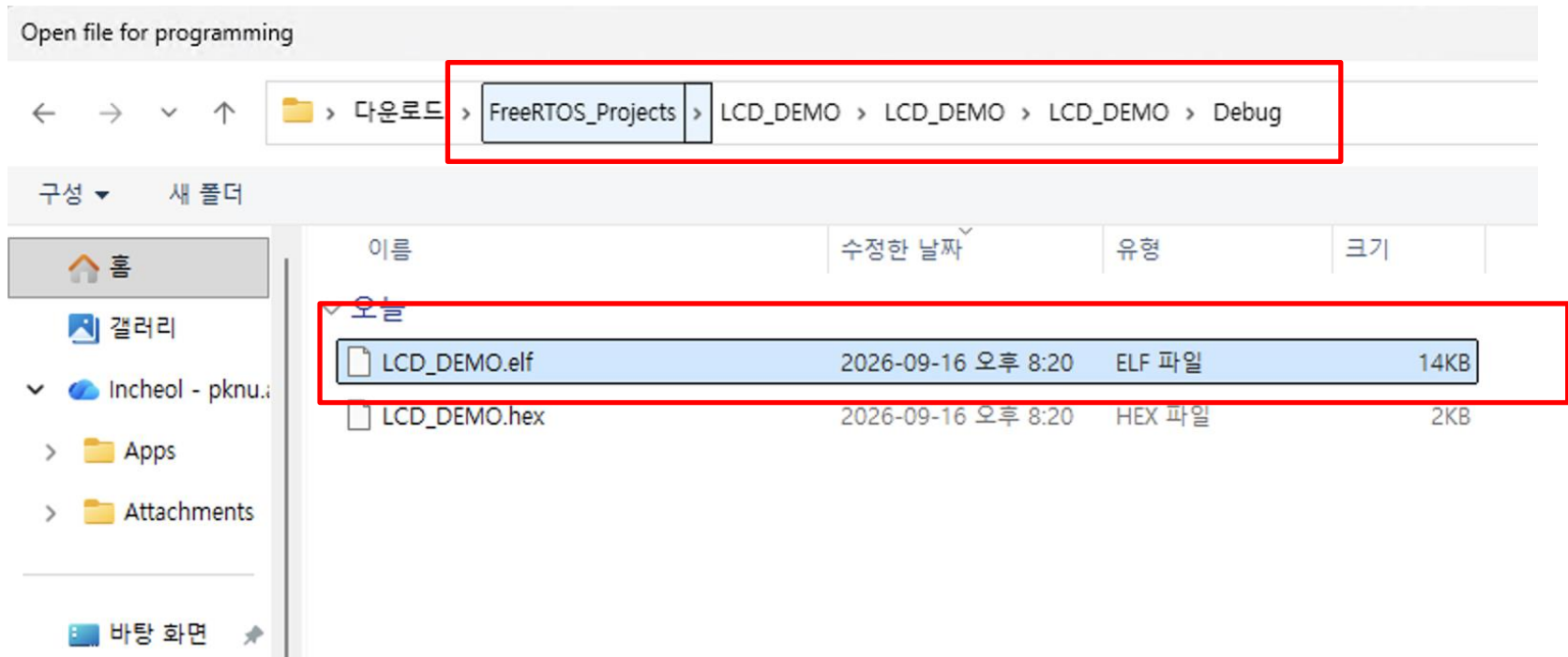
Starting operation read registers
Reading register EXTENDED...OK
Reading register HIGH...OK
Reading register LOW...OK
Read registers...OK

Read registers...OK

Close

LCD Demo 실습

- Device Programming (계속)
 - LCD_DEMO.elf 실행파일 선택



LCD Demo 실행

AVRISP mkII (000200212345) - Device Programming

Tool: AVRISP mkII | Device: ATmega128 | Interface: ISP | Device signature: 0x1E9702 | Target Voltage: 3.3 V

Interface settings
Tool information
Device information
Oscillator calibration
Memories
Fuses
Lock bits
Production file

Device
Erase Chip | Erase now

Flash (128 KB)
C:\Users\IncheolShin\Downloads\FreeRTOS_Projects\LCD_DEMO\LCD_DEMO\LCD_DEMO...
☒ Erase device before programming
☒ Verify Flash after programming
☐ Advanced
Program | Verify | Read...

EEPROM (4 KB)
...
☒ Verify EEPROM after programming
☐ Advanced
Program | Verify | Read...

Starting operation read registers
Reading register EXTENDED...OK
Reading register HIGH...OK
Reading register LOW...OK
Read registers...OK

Read registers...OK

Close